AHMED H. HANFY

Aerodynamic specialist | Scientific computing Engineer | Mechanical Engineer

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Summary

Doctoral Researcher enrolled in the Marie Curie fellowship program, with specialized experience in experimental aerodynamics. Holds Master's in Applied Mathematics and Scientific Computing, with prior experience as a junior mechanical designer. Adept in various programming languages such as Python, MATLAB, and C++; with a focus on data and image analysis. Proficient in CAD modeling using Autodesk Inventor and familiar with Siemens NX. Seeking to contribute expertise in aerodynamics and pursue professional growth.

WORK EXPERIENCE

11/2020- Present

Experimental Aerodynamics Specialist

Institute of Fluid-Flow Machinery, Polish Academy of Sciences (IMP PAN), Poland

- Expert in transonic testing, and advanced data analysis. Actively contributed to EU research projects in collaboration with major industry partners.
- Core Competencies: Wind tunnel operation, Schlieren (high-speed, multi-color), BOS, LDA, oil visualization.
- Data Analysis & Software: Python tools for image/signal processing, statistics, boundary layer, isentropic flow, shock tracking.
- Instrumentation & Control: LabVIEW/DAQs for feedback and camera triggering (±0.3ms trigger accuracy); flow control (suction, Reynolds).

Project Experience:

11/2020-07/2021 — SMS Project:
- Performed POD analysis on transonic PIV data for noise reduction.
- Analysed the impact of vibrating trailing edges on shock wave unsteadiness.

08/2021-present — TEAMAero Project:
- Led test planning for fan profile surface roughness studies.
- Coordinated closely with Rolls Royce Deutschland for the application of surface textures on fan profiles.
- The study revealed a reduction in separation by 1%, and momentum thickness by 10% but increase in shock unsteadiness by 23%.
- Implemented an inlet valve for Reynolds number control in stator profile testing.

03/2018- 09/2018

Junior Mechanical Designer

Taqat ME (Renewables and Environment), Egypt

- Designed and modeled sheet metal solar tanks with various sizes.
- Enhanced the water heating process using coil fins by 10%.
- Document the manufacturing process and plan inspection procedures.
- Supervise the manufacturing and inspection of solar tanks.
- Design and supervise the construction of a 20k-liter water tank.

INTERNSHIPS & SCIENTIFIC VISITS

08/2022- 09/2022

Visiting Researcher

German Aerospace Centre (DLR), Cologne (Germany)

Within Marie Skłodowska-Curie Actions, a scientific secondment was made to DLR Transonic Cascade Wind Tunnel for unsteady measurements, such as:
- Calibration of Kulite probe and vibration sensors.
- Apply analysis on high Data Acquisition rates up to 200KHz.
- Synchronized speed Schlieren setup.

SKILLS

LANGUAGES SKILLS

English (business proficiency)

Polish

Arabic (Native)

SOFTWARE SKILLS

Inventor
AutoCAD
Siemens NX

MATLAB
Ansys (Fluent)
LabVIEW

Python
(OpenCV, SciPy, Pandas, etc.)
C++
(Vector, OpenMPI, etc.)
Fortran

Photoshop
3D Max

Other skills

3D printing | Manufacturing | Pneumatics circuits | FEA | Mechanical Drawing | Robotics | MATLAB GUI | C# | JavaScript | Analysis | PCA | Dataset Analysis | Montecarlo simulation | OOP | Heat transfer | Microsoft Office | Linux |

02/2020-05/2020

Research Internship

Institute of Fluid-Flow Machinery, Polish Academy of Sciences (IMP PAN), Poland

As a complementary course during master's study, with these objectives:

- Extracts quantitative information from interferograms with uncertainty less than 2% for experimental study relevant to SWBLI.
- Developing MATLAB desktop app involving Fast Fourier Transform (FFT) and phase shifting.
- Improve phase detection accuracy using machine learning techniques (DBSCAN, Linear regression, etc.).

08/2013-08/2015

Undergraduate summer trainings

- Hydro-electric stations training centre, Aswan (Egypt).
- MANTRAC (Caterpillar Dealer), Alexandria (Egypt).
- Ansys Fluent Workshop, ASME Alexandria chapter, (Egypt).

EDUCATION

11/2020– Present (Est Graduation, 2025)

Doctor of Philosophy (Ph.D.), Mechanical engineering

Institute of Fluid-Flow Machinery, Polish Academy of Sciences (IMP PAN), Poland

Dissertation: *"Surface roughness effect on SBLI on transonic compressor profile"*

Relevant coursework:

- *SWBLI and Flow Control (University of Cambridge),*
- *Recent Development in CFD (University of Glasgow and Cadence "NUMECA"),*
- *Experimental Methods in Flow Field Diagnostic (AMU, TU Delft and ONERA),*
- *Advanced Particle Image Velocimetry measurements (TU Delft),*
- *Aerothermal Methods for Design of Turbomachinery Components and related CFD approaches (TU Berlin, Rolls Royce Deutschland).*

Recent Publications:

- Hanfy, A. H., Flaszynski, P., Doerffer, P., & Kaczynski, P. (2026). *Enhanced line-scanning method for shock wave tracking in high-speed schlieren imaging.* Aerospace Science and Technology, 111022. DOI: 10.1016/j.ast.2025.111022
- Hanfy, A.H., Flaszyński, P., Kaczyński, P., Doerffer, P. (2025). *Surface Roughness Effect on Shock Boundary Layer Interaction on Compressor Rotor Profile.* In: Flaszynski, P. et al (eds). Towards Effective Flow Control and Mitigation of Shock Effects in Aeronautical Applications. vol 201. Springer, (P. 325–344). DOI: 10.1007/978-3-031-86605-0_15
- Nel, P. L., Janke, C., Vasilopoulos, I., Swoboda, M., Schreyer, A.-M., Hady, A., & Flaszyński, P. (2024). *Effect of transition on self-sustained shock oscillations in highly loaded transonic rotors.* AIAA Journal, 0(0):1–13. DOI: 10.2514/1.J063378

Recent conferences:

03/2025 "*Experimental Studies of Shock Wave Boundary Layer Interaction on Suction Side of Fan Representative Profile*",
16th European Conference on Turbomachinery Fluid Dynamics (ETC16), Hannover, Germany

09/2023 "*Surface roughness effect on boundary layer and shock-induced separation on transonic compressor profile*",
2nd Colloquium on Separation Control in High-Speed Flows – mechanisms, methods, and application, Aachen, Germany

09/2018-05/2020

Master of Science (M.Sc.), Mathematical Engineering and Applied Mathematics

InterMaths Joint MSc (Double Degree) Programme

09/2019-05/2020 Ivan Franko National University of Lviv (IFNUL), Ukraine

Relevant coursework:

Optimization of complex systems; Mathematical modelling and simulation; Algorithms and data structure.

09/2018-07/2019 University of L'Aquila (UAQ), Italy

Relevant coursework:

Parallel Computing, Numerical Methods for Linear Algebra and Optimization, Stochastic Modelling and Simulations, Data Analytics, Big Data.

10/2011-07/2016

Bachelor of Science (B.Sc.), Mechanical Engineering

Alexandria University (AU), Egypt

Relevant coursework:

Mathematics; Physics; Fluid mechanics I-II; Gas dynamics; Thermodynamics I-II; Heat transfer; Technical writing; Mechanical design I-II-III.

PROJECTS

- **Shock Tracking Library**
  [12/2022-05/2024]
  Schlieren image analysis, (OOP, OpenCV, Python Package).
- **Finite Fringe Analysis for Optical Measurement of Compressible Fluid Flow Parameters**
  [02/2020-05/2020]
  Automated image analysis using FFT (MATLAB) - MSc. Thesis.
- **Parallel implementation of Poisson's equation**
  [05/2019]
  Final project for the parallel computing course (OpenMPI Fortran and C++)
- **Machine learning for hydraulic condition monitoring systems**
  [06/2019]
  Final project for Data Analysis & Big Data courses in (Python)
- **CFD applications in oil and gas industry**
  [10/2015-07/2016]
  CFD Multiphase (Ansys Fluent) – BSc. Project

ACTIVITIES

- **Researchers' night in Poland**
  [09/2021-10/2022]
  As a speaker to encourage junior researchers.
- **Volunteer at Science Club team**
  [05/2012-06/2016]
  Positions: Co-founder, Media member Graphic designer & Team leader.
- **Volunteer at Torpedo robotics team** (student organization)
  [02/2014-06/2016]
  Participate in the MATE ROV competition to build remotely operated vehicles (ROV) that perform underwater tasks. Roles: Head of Mechanical sub-team. Technical asset and review outputs. Boost designs using (FEA, CFD).

REFERENCES

- **Prof. Pawel Flaszynski**
  Head of Aerodynamics Department, Institute of fluid-flow machinery polish academy of sciences, Poland
  PhD supervisor, pflaszyn@imp.gda.pl
- **PhD. Eng. Janusz Telega**
  Experimental Aerodynamics Department, Institute of fluid-flow machinery polish academy of sciences, Poland
  M.Sc. co-advisor, januszt@imp.gda.pl
- **Assoc. Prof. Yuriy Yashchuk**
  Faculty of Applied Mathematics and Informatics, Ivan Franko National University of Lviv, Ukraine.
  M.Sc. supervisor, yuriy.yashchuk@lnu.edu.ua
- **Prof. Kamel Elshorbagy**
  Full professor of Fluid mechanics, Mechanical Eng. Dept. Alexandria University, Egypt.
  Torpedo team supervisor, kshorbagy@yahoo.com